Pre-registration template for secondary economic data

This template is described and associated to the paper “Pre-registration for Economists: exhaustive templates for primary and secondary data”, for which the working paper version is accessible here: *https://hal.science/hal-04943071.*

Researchers will find below the detailed template for all the elements that should be included in a pre-registration for secondary data. They should simply follow the steps and adapt each category to their research project specificities. Importantly, most categories include a justification box. This box allows researchers to justify why they have omitted certain elements and discuss how this affects the severity of testing. The justifications, as all other elements in this template, are optional. However, omitting elements and failing to justify increases the ex-post analysis space. It is then up to editors, reviewers, and readers to evaluate the impact on the severity of testing.

Preregistering secondary data can be tricky, especially when researchers already have access to the data. In this case, researchers should refer to Section 4 in the associated paper to determine whether a preregistration is relevant. If it is not relevant, researchers should be transparent and explicitly mention that their research project is exploratory. Exploratory research is a cornerstone and not preregistering does not decrease a project's quality (see Aczel(2024) for more details on exploratory research).

Administrative information

Before accessing the data or undergoing any form of analysis, researchers should anticipate administrative information surrounding a research project. This information is important as it allows editors, reviewers, and readers to quickly identify authors' pre-defined roles and data structure. Failure to report some of the elements can increase analysis space and should be justified.	
Does the data already exist?	✓ Yes O No *If no, the researchers should refer to the primary data pre-registration template.*
Do the researchers describe the type of data to be used? *Researchers should provide an overview of the type of data to be used, focusing on the data source, data availability, and data confidentiality. Researchers do not need to be too specific here but should provide an overview of the data type. Researchers will have the opportunity to accurately describe the data to be used in the Sampling plan section below.*	✓ Yes *If yes, please describe the type of data:* The data is from the database SOEP (https://companion.soep.de/) through the #ManyDaughters project (https://www.manydaughters.com/). The access was restricted to the datasets biobirth, biol, bioparen, hgen, hl, pgen, pl, and ppathl. The restriction was applied as the datasets are part of the metascience project #ManyDaughters. The datasets were provided in .dta format and only 5% of the total dataset was accessible.

	O No *If no, researchers do not include a description of the type of data.* Justification:
Do the researchers list the plan contributions for each co-author? *In a transparency effort, researchers can provide a list of the contributions made by each contributor. This list can be added in the pre-registration or in the final manuscript. However, listing the researchers' contributions at the pre-registration stage clarifies who drafted the pre-registration. For example, Nature journals recommend the CRediT designation available here: https://credit.niso.org/.*	✓ Yes If yes, please enter each author's contribution: Matthieu Pourieux and Thibaut Arpinon will both equally contribute to the project conceptualization, PAP drafting, coding, and formal analysis. Due to the specificity of the metascience project, a CRediT contribution could not be used. *If you would like to enter CRediT designation, please leave this space empty and proceed below.* O No *If no, Researchers have decided not to include authors' contributions.* Justification:
Would you like to list the contributions of each author according to the CRediT designation? *For each author, you can click on the contributions.*	O Yes *If yes, please add the authors below and their CRediT contributions.* ✓ No Add author *If author is added then:* Author 1: O Conceptualization O Data curation O Formal analysis O Funding acquisition O Investigation O Methodology O Project administration O Software O Resources O Supervision O Validation O Visualization O Writing – original draft O Writing – review & editing

Do the researchers designate a Research Guarantor? *Researchers can select one researcher among the consortium who will take on the responsibility of the research integrity. For more information please see Rennie et al.(1997); Sánchez-Jiménez et al.(2017)*	O Yes *If yes, write name of Research Guarantor* Research Guarantor: ✓ No *If no, Researchers do not to designate a Research Guarantor.* Justification: The project is not compatible with a research guarantor.
Do the researchers provide an access link to the dataset?	O Yes *If yes, please copy a link to access the data:* ✓ No *If no, researchers do not provide a link to access the data.* Justification: We cannot provide an access link because the access was provided by the #ManyDaughters project.
Do the researchers provide ethics and/or IRB approval?	O Yes *If yes, please copy a link to the ethics or IRB approval:* ✓ No *If no, researchers do not include a copy of the ethics or IRB approval.* Justification: We do not provide an IRB or ethics approval as the project is coordinated by the #ManyDaughters team.

Do the researcher provide information about the project's funding?	O Yes *If yes, please detail the funding source and any relevant information:* ✓ No *If no, researchers state that the project did not receive funding, or they do not wish to share funding information.* Justification:
Do the researchers provide a general description of their project?	O Yes *If yes, please enter a project description:* ✓ No *If no, researchers do not to include a project description.* Justification: We do not provide a project description as the project is coordinated by the #ManyDaughters team.
Do the researchers provide a project timeline?	O Yes *If yes, please enter the project timeline:* ✓ No *If no, researchers do not to include a project timeline.* Justification: We do not a project timeline as the project is coordinated by the #ManyDaughters team.

Are there any additional administrative information that the researchers would like to include in the pre-registration?	O Yes *Please enter any additional relevant elements:* ✓ No *If no, researchers have decided not to include any additional elements.*

Sampling plan

Description of the data *When undertaking a pre-registration for secondary data, researchers should have a clear idea of the data to be used or accessed. This is important for transparency and for future potential replications. The more details researchers provide on the data, the lower the likelihood of undisclosed data manipulations and the easier it will be for future research projects to replicate the data. However, unanticipated events (e.g., data structure is different than described) can occur and alter the final dataset. In this case, researchers have two options. First, if the alteration occurs before the pre-registration is submitted, researchers should modify the pre-registration ex-ante rather than attempt to rationalize the changes ex-post and transparently report it. Second, if the alteration occurs once the pre-registration has been submitted, the researchers should report the changes transparently in their final manuscript. Researchers should also discuss how these changes affected the pre-registered plan but also discuss whether this decreased or increased the severity of the testing. For more elements on deviations, please refer to the template "Deviations from pre-registration".*	
Do the researchers indicate where the data is located? *Researchers should detail, whenever possible, the data source, where it is located, and for public databases, how other researchers can access it.* *Note of caution: In case the data is confidential, not public or cannot be shared, researchers should provide as much detail as possible without compromising confidentiality.*	✓ Yes *If yes, please describe where the data is located. Researchers should provide a concise description.* User-friendly documentation about the data can be on the SOEP companion (https://companion.soep.de/) and access was provided by the #Manydaughters team. O No *If no, researchers do not describe where the data is located.* Justification:
Do the researchers indicate when the data will be accessed? *For public databases, researchers should mention that they had access to the data before the pre-registration. For confidential data or guarded data, researchers should provide a time-stamp for when the data access will occur. This can be a fixed date or a timespan. If the data access dates change due to unpredicted events, researchers should report it in their final manuscripts.* *Note of caution: Ideally, the data access should be dated after the pre-registration submission date. If the data access is dated before, the researchers' degree of freedom and the risk of Pre-registering After the Results are Known (PARKing) greatly increase.*	✓ Yes *If yes, please describe when the data will be accessed. Researchers should provide a concise description.* The data was first accessed in May 2025. As part of the #ManyDaughters project, only 5% of the total dataset was accessible. O No *If no, researchers do not describe when the data will be accessed.* Justification:

	If yes, do the researchers provide a document from a gatekeeper proving that the data was not accessed previously? ✓ Yes *If yes, please provide a link to the gatekeeper document.* The data was provided by the #ManyDaugthers team, who served as a gatekeeper. O No *If no, researchers do not provide a document from a gatekeeper.* Justification:
For representative samples, do researchers pre-specify which quotas/categories will be used to ensure the representativeness? *Researchers should pre-specify which quotas/categories they will use (e.g., age, gender, and area of living).*	O The data will be representative *If yes, please describe which quotas/categories will be used to ensure the representativeness.* ✓ The data will not be representative *If no, researchers state that the data will not be representative.*
Do the researchers describe the data? *Researchers should provide detail, whenever possible, on two data aspects. First, researchers should describe the type of data (cross-sectional, time series, panel, or any other type). Second, researchers should provide observation-related detail. For individual data, this could involve specific individual characteristics (e.g., location in countries, regions, villages). For aggregate or indicator data, this could involve the observations timeframe or calculation methods.*	✓ Yes *If yes, please describe the data. Researchers should provide a concise description.* Even though the SOEP data is longitudinal, we focus only on the data from 2019 and therefore retain a cross-sectional dataset across all hypotheses. We do so because our variables of interest are only available for 2019. However, we modify the data according to our hypotheses. For hypotheses H1, H2, and H4, we use individual-level data. For H3, we aggregate to use household-level data. O No *If no, researchers do not describe the data.* Justification:

Do the researchers mention whether multiple datasets will be combined? *In some cases, researchers combine datasets from multiple sources. Researchers should provide detail on all datasets above and should mention how the datasets are going to be merged together.*	✓ Yes *If yes, researchers combine multiple datasets.* O No *If no, the researchers do not combine multiple datasets.*
	If yes, do the researchers provide the combination method? ✓ Yes *If yes, please describe how the datasets are combined.* The SOEP datasets biobirth, biol, bioparen, hgen, hl, pgen, pl, and ppathl are combined. O No *If no, the researchers do not describe how the datasets are combined.* Justification:

Sample size rationale

For secondary data, the sample size rationale is often constrained by the number of observations available in a dataset. Defining a sample size rationale can come in handy when the researchers wish to justify access to a certain number of observations or to calculate a required effect size to reach enough power. Researchers have different ways to define the sample size. A Power Analysis using a-priori power is the most accepted practice for studies aiming to reject a null hypothesis using a predefined alpha level. In case this approach is not feasible, researchers have different options listed below. For additional information, researchers can refer to Lakens (2022) and Arpinon and Espinosa (2023).

Do the researchers provide a justification or rationale for the choice of the sample size?	O Yes *If yes is selected, see elements in the right column.* ✓ No *If no is selected, researchers do not provide a justification or rationale for the sample size.* Justification: The sample size is constrained by the #ManyDaughters team. ***If no, skip to the next question in the left column and leave the right column empty.***	**Which type of justification is provided by the researchers?** *Researchers can rely on the following approaches to calculate the a-priori power:* • *Smallest Effect Size of Interest (SESOI):* researchers estimate the number of observations necessary to detect a pre-defined effect size, given an alpha and beta level. • *Minimum Detectable Effect (MDE):* researchers fix the sample size and estimate the level of power based on different effect sizes, for a given alpha. • *Sensitivity analysis:* researchers estimate the effect size necessary to detect a significant effect given a sample size, alpha level, and level of power. • *Confidence interval of the expected effect:* researchers can define a range for the confidence interval of the expected effect size to fall into and exclude values outside of this range. • *Expected Effect Size:* researchers define an effect size from a theoretical prediction, a meta-analytic effect size, or a previous estimate in the literature. However, researchers should be aware that in the presence of publication bias, effect sizes in meta-analyses and in the literature can be inflated. Anderson et al. (2017) discuss this issue and how to correct for it. • *Distribution of effect sizes:* researchers have the distribution of effect sizes in a particular research area and calculate the number of observations necessary to detect, given an alpha and beta level. *Note of caution: Researchers should avoid calibrating power analyses based on effect sizes from pilot studies. Albers and Lakens (2018) discuss this issue.* *O Smallest Effect Size of Interest (SESOI)* *O Minimum Detectable Effect (MDE)* *O Sensitivity analysis* *O Confidence interval of the expected effect* *O Expected Effect Size* *O Distribution of effect sizes* *O Other:*

		Do the researchers discuss an effect that is economically relevant? *As economists, researchers should not simply consider calibrating their power analyses using an effect size different from zero, but an effect size that is economically relevant (e.g., using a cost-benefit analysis). Arpinon and Espinosa (2023) discuss this approach.* O Yes *If yes, please explain how the effect is economically relevant.* O No *If no, researchers do not discuss an economically relevant effect size.* Justification:
		Do the researchers use a Multiple Hypothesis Testing (MHT) correction in the power calculation? *When calculating power, researchers should anticipate whether they will use a MHT correction, as this procedure can affect the power-level. The MHT should be the same as described in the Analysis Plan section (see section below for more detail). However, in some cases, researchers can use an MHT correction that is more severe in the power analysis to ensure maximal power.* O Yes *If yes, please describe the MHT correction used. Researchers should provide a concise description.* O No *If no, researchers do not use an MHT correction in the power calculation.* Justification:

		Is the Alpha level reported? *The common alpha levels reported in economics are 0.1, 0.05, and 0.01. The commonly accepted level is 0.05. However, researchers should keep in mind that some journals may require specific alpha levels.* O Yes *If yes, please provide the alpha level and reason for choosing this level. Researchers should provide a concise description.* O No *If no, researchers do not pre-specify an alpha level.* Justification:
		Do the researchers provide the power analysis results? O Yes *If yes, please provide the power analysis results. Researchers should provide concise and accurate results.* O No *If no, researchers do not provide the power analysis results.* Justification:
		Do the researchers provide the power analysis code? O Yes

		If yes, which software is used for the power analysis coding? If yes, please copy a link to the power analysis code: O No *If no, researchers do not provide the power analysis code.* Justification:
Are the researchers attempting to reach a degree of precision (i.e., specific width of the confidence interval around the effect size)? *This question should be treated if the answer to the previous question is No.*	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers are not attempting to reach a degree of precision.* Justification: The sample size is constrained by the #ManyDaughters team. ***If no, skip to the next question in the left column and leave the right column empty.***	**Do the researchers provide a justification for the expected effect size?** O Yes *If yes, please provide an expected effect size justification. Researchers should provide a concise description.* O No *If no, researchers do not provide an expected effect size justification.* Justification:

		Do the researchers provide a justification for the assumption of the population standard deviation? O Yes *If yes, please provide a justification for the assumption of the population standard deviation. Researchers should provide a concise description.* O No *If no, researchers do not provide a justification for the assumption of the population standard deviation.* Justification:
		Do the researchers provide detailed calculations? O Yes *If yes, please provide a justification for the detailed calculations. Researchers should provide a concise description.* O No *If no, researchers do not provide detailed calculations.* Justification:

Do the researchers face resource limitations that constrain the sample size? *This question should be treated if the answer to the previous question is No.*	✓ Yes *If yes is selected, see elements in right column.* O No *If no is selected, researchers are not constrained by the sample size.* Justification: ***If no, skip to the next question in the left column and leave the right column empty.***	**Are the researchers able to capture an entire population?** O Yes ✓ No *If no, do the researchers use an a-priori power analysis to estimate and discuss the level of power for a given effect size and constrained sample size? Alternatively, do the researchers estimate and discuss the required effect size needed to attain a certain level of power with the limited sample size?* Justification:

Do the researchers define the sample size using heuristics or commonly accepted practices described by the literature? *This question should be treated if the answer to the previous question is No.*	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers do not define the sample size using heuristics or commonly accepted practices described by the literature.* Justification: The sample size is constrained by the #ManyDaughters team. ***If no, skip to the next section and leave the right column empty.***	**If yes, do the researchers accurately describe the sources, intuitions, and logic of the heuristics or commonly accepted practice to define the sample size?** *Note of caution:* • *Sample size justifications based on previous papers sample sizes are likely to be biased due to the presence of publication bias.* • *Researchers using a sample size justification based on heuristics or previous sample size collections need to provide a thorough justification, as cherry-picking sample sizes to accommodate the researchers needs is likely.* O Yes *If yes, please describe the sources, intuitions, and logic of the heuristics. . Researchers should provide a concise description.* O No *If no, researchers do not describe the sources, intuitions, and logic of the heuristics.* Justification:

Outcomes of interest

Researchers should anticipate the main outcome variables and be able to accurately describe them. When defining primary vs. secondary outcomes, researchers should anticipate the general goal of their analysis and how this will affect the quality of the evidence. Researchers should then anticipate what elements will be part of the confirmatory analysis and what elements will be part of the exploratory analysis, bearing in mind how this affects the severity of testing. As mentioned earlier, researchers might run into unanticipated issues when accessing the data that force deviations from the original pre-registration. In this case, the changes should be reported as deviations and a discussion on how this impacts the severity of testing should be provided. For more elements on deviations, please refer to the template "Deviations from the preregistration".

Do the researchers provide their primary outcome(s)? *The researchers should provide a main variable or set of main variables that will be analyzed as part of the confirmatory analysis.* *Note of caution: Researchers should keep in mind that the inclusion of an additional primary outcome increases the analysis space. This should then be considered using a Multiple Hypothesis Testing correction. This point is discussed below in the section "Analysis Plan".*	✓ Yes *If yes is selected, see elements in right column.* O No *If no is selected, researchers do not provide their primary outcome(s).* Justification:	**If yes, do the authors describe the construction of the primary outcome(s)?** *Researchers should provide an accurate description of the main variable(s). The main variable(s) can be taken directly from the data (e.g., amount given to a charity). However, sometimes researchers may need to transform the main variable(s) to fit the data analysis needs. In this case, researchers should describe how the transformation will occur.* *Note of caution: Not reporting a variable transformation in the pre-registration can be costly as this increases the analysis space and decreases the severity of testing. If the researchers forget to include this element, they should report it as a deviation (please refer to the template "Deviations from the pre-registration)".* *If yes, please provide the primary outcome(s) description.* For H1, the primary outcome is a composite score from the variables ph0395i01 to ph0395i06 in the combined dataset. These variables are Likert scale responses ranging from 1 (*Do not agree at all*) to 7 (*Completely agree*). Participants respond for the following statements: – *It's good that legislators take into account that not everyone can be classified as either male or female* – *It should be accepted in our society that not everyone identifies with their biological sex* – *It's completely natural that some people are born with both male and female biological features* – *Sex assignment is not so important for a person's development* – *Schools and preschools should promote non-gender-typical activities that are not typical for children's gender* – *It should be accepted in our society that same-sex couples want to have children* To compute the individual score, we average for each respondent the raw values of the 6 statements, normalize the value (i.e., score between 0 and 1) and name the variable dep_var_h1. For H2, the primary outcome is the variable plh004 in the combined dataset. This variable ranges from 0 (*Far left*) to 10 (*Far right*). Participants respond to the following question: *In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?* We reverse the score to fit the hypothesis (10 = far left; 0 = far right) and name the variable dep_var_h2.

		For H3, we aggregate, for each individual, the reported hours spent on a typical weekday, Saturday, and Sunday what we consider "women-specific stereotypical activities": • Time on childcare (variables pli0019_h, pli0022_h, pli0044_h) • Time on housework (variables pli0012_h, pli0016_h, pli0043_h) • Time on errands (variables pli0011 pli0054, pli0040) • Time on support of persons (variables pli0057, pli0055, pli0046) We aggregate, for each household (i.e., by household id number hid), the total number of hours dedicated to those activities. The primary outcome is the share of hours done by the man in the household (i.e., variable ranging between 0 and 1). Note that for households amounting to 0 hours, we attributed an equal share between men and women (i.e., 0.5). We name the variable dep_var_h3. For H4, the primary outcome is identical as the primary outcome for H1 (see above) and the variable is named dep_var_h4. For all hypotheses, we use a dummy variable to capture participants who have a daughter (i.e., variable is equal to 1 if the participant has a daughter). We name the variable d_daughter.
Do the researchers provide their secondary outcome(s)? *The researchers can provide a secondary variable or set of secondary variables that will be analyzed as part of the exploratory analysis. Anticipating potential secondary outcomes and describing them can increase the severity of the testing on said outcomes.*	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers do not provide their secondary outcome.* Justification: The project #ManyDaughters does not allow for secondary outcomes.	**If yes, do the authors describe the construction of the second outcome(s)?** *If yes, please describe the secondary outcome(s) description:*

Does randomization occur in the data? *In some cases, participants are randomized into conditions or blocks. This should be described.*	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers state that randomization does not occur in the data.* Justification: No randomization occurs in the data, and we do not use additional ex-post statistical techniques to assign participants into conditions.	**If yes, do the authors describe how the randomization occurs?** *Researchers should accurately describe how the randomization will occur, who will be randomized, and the randomization level. This includes describing the randomization method (e.g., matching).* O Yes *If yes, please describe how the randomization occurs.* O No *If no, researchers do not describe how the randomization will occur.* Justification:
Will blinding occur? *Blinding refers here to the level of knowledge about the data setup. Blinding occurs when at least one of the parties (participants, researchers, or persons collecting the data) is unaware of one or more data structure characteristics.*	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers state that blinding will not occur.* Justification: No blinding occurs as we are constrained by the #Manydaughters project.	**If yes, do the researcher describe how the blinding will occur?** *Researchers should describe accurately who will be blinded. Single-blinded studies refer to situations where the participants do not know they are part of a condition, but the researchers know. Double-blinded studies refer to neither the participants nor the researchers knowing who is part of a condition.* O Yes *If yes, please describe how the blinding will occur.* O No *If no, researchers do not describe how the blinding will occur.* Justification:

Other project specific elements *Researchers may be using unique features that do not fit in any of the Sampling Plan categories. Researchers should describe such features as precisely as possible. Omitting to do so may cause confusion and be interpreted as an increase in analysis space.*	
Do the researchers have any additional features that do not fit in the categories above?	O Yes *If yes, please describe the additional features.* ✓ No *If no, researchers do not have any additional features.*

Analysis plan

<table>
<tr><td colspan="3">Inclusion and exclusion rules
At the pre-registration stage, researchers should anticipate potential observations in the data that will not be included in the final analysis and provide exclusion rules/inclusion rules. These rules need to be pre-specified to limit the analysis space. As mentioned earlier, researchers might run into unanticipated issues when accessing the data, forcing deviations from the pre-registered exclusion rules. In this case, the changes should be reported as deviations and a discussion on how this impacts the severity of testing should be provided. For more elements on deviations, please refer to the template “Deviations from the pre-registration”.</td></tr>
<tr><td>Do the researchers provide exclusion rules?
Researchers should provide a set of pre-specified exclusion rules for observations that will not be included in the final analysis.</td><td>✓ Yes
If yes is selected, see elements in right column.

O No
If no is selected, researchers do not provide exclusion rules.
Justification:</td><td>If yes, do the researchers describe which observations will be excluded from the final analysis?
Researchers should describe as accurately as possible the combination of variables and observations that will not be included in the final analysis. This can include, but is not limited to, the following:
Missing observations
Erroneous observations
Inconsistent or overly consistent values
Observations with values above/below a predefined threshold
Suspicious responses (e.g., bot generated)
Individual-specific characteristics (e.g., demographics)
Methodology based rules (e.g., inconsistent preferences)
Equipment errors
Any other data-specific occurrences
<u>Note of caution:</u> Researchers should attempt to anticipate as many exclusion rules as possible. Any exclusion rule that is not pre-registered but added to the final analysis increases the analysis space and decreases the severity of testing. In this case, the deviations should be reported and a discussion of whether they affect the severity of testing should be provided.

✓ Yes
If yes, please describe the exclusion rules:
For all hypotheses, we apply three exclusion rules:
We restrict the sample to the year 2019 (variable syear = 2019).
We keep only participants for which the biographical information is sufficient (variable bioinfo = 1, 2, 3, or 4).
We keep only participants who have kids (variable sumkids > 0).</td></tr>
</table>

		We then apply specific exclusion rules for each hypothesis. For H1, we exclude all observations that have not responded to at least one of the gender attitudes questions (i.e., missing values for variables ph0395i01 to ph0395i06). For H2, we exclude all observations that have not responded to the political orientation question (i.e., missing value for variable plh0004). For H3, we restrict the sample to household ids with two respondents (i.e., variable hids is present twice). We then restrict to households in which the two respondents are one male and one female (i.e., variable sex = 1 and 2) and who reported having a partner (i.e., variable partner = 1, 2, or 3). We also exclude all participants who did not report, reported implausible values, or were not included in the version of the questionnaire for the women-specific stereotypical activities (i.e., observations missing or equal to -2, -3, or -5 in variables pli0019_h, pli0022_h, pli0044_h, pli0012_h, pli0016_h, pli0043_h, pli0011 pli0054, pli0040, pli0057, pli0055, and pli0046). For H4, we exclude all observations that have not responded to at least one of the gender attitudes questions (i.e., missing values for variables ph0395i01 to ph0395i06). O No *If no, the researchers do not describe the exclusion rules.* Justification**:**
Do the researchers provide a correction method for missing/outlier values?	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers do correct for missing/outlier values.* Justification: We delete all missing observations from the database.	**If yes, do the authors describe how they will correct for missing/outlier values?** *Researchers who would like to include observations that have missing/outlier values should describe the predefined correction method. Some examples include, but are not limited, to the following:* • Winzorisation • Mean imputation • Factor-based imputation *Some resources for dealing with outliers: Beaumont and Rivest(2009); Aguinis et al.(2013); Leys et al. (2019); Arpinon and Espinosa (2023)*

		O Yes *If yes, please describe the correction method:* O No *If no, researchers do not describe the correction method for missing/outlier values.* Justification:

Hypotheses / Questions of interest *Researchers should provide their main questions of interests/hypotheses at the pre-registration stage to avoid the risk of Hypothesizing After the Results are Known (HARKing). Two possible situations can arise. First, researchers may be testing hypotheses that they derive from a theoretical prediction or from the literature. In this case, they should clearly state the hypotheses to be tested and, whenever possible, provide a directional hypothesis. Second, researchers might be conducting original research that is not theory-based nor been studied in the literature, and for which they do not anticipate a clear effect. In this case, they should mention clearly that the analysis is non-hypothesis driven.* *Researchers should attempt to describe as precisely as possible the questions that will be explored (e.g., exploring the effect of Z and W on Y). As mentioned previously, any research question or hypothesis added ex-post increases the analysis space and decreases the severity of testing. Any question of interest added ex-post should be interpreted as exploratory results (or lacking sufficient proof) and be reported as a deviation. For more elements on deviations, please refer to the template "Deviations from the pre-registration".*		
Do the researchers specify their hypotheses?	✓ Yes *If yes is selected, see elements in right column.* O No *If no is selected, researchers do not specify hypotheses.* Justification:	**If yes, do the authors clearly state the relationships between the variables?** ✓ Yes *If yes, please describe the relationships between the variables.* For this project, we test four hypotheses provided by the #Manydaughters team: *H1. Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for respondents with daughters)> (X for respondents without daughters).* Specifically, we will test whether having a daughter increases the score on the gender equality scale (see calculation in the primary outcome section above).

		H2. Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters). Specifically, we will test whether having a daughter increases the score on the political orientation scale (see calculation in the primary outcome section above). *H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters)> (Z for respondents without daughters).* Specifically, we will test whether having a daughter increases the share of women-specific stereotypical activities done by men within the household (see calculation in the primary outcome section above). *H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality, X for male respondents with daughters − X for male respondents without daughters > X for female respondents with daughters − X for female respondents without daughters.* Specifically, we will test whether men having a daughter increase their score on the gender equality scale more than women (see calculation in the primary outcome section above). O No *If no, researchers do not describe the relationships between the variables.* Justification:

		If yes, do the authors provide a directional hypothesis? ✓ Yes *If yes, please describe the direction of the hypothesis.* All hypotheses provided by the #ManyDaughters team are directional hypotheses (see above). O No *If no, researchers do not describe direction of the hypothesis.* Justification:

Statistical testing

Researchers should clarify the statistical methodology that they will use to test their hypotheses. This should be as detailed as possible (e.g., covariates, robust standard errors, Lasso, etc.). As mentioned previously, if the researchers decide to change the statistical methodology ex-post, this should be clearly reported as a deviation. For more elements on deviations, please refer to the template "Deviations from the pre-registration".

Do the researchers provide their statistical testing methodology?	✓ Yes *If yes is selected, see elements in right column.* O No *If no is selected, researchers do not specify their statistical testing methodology.* Justification:	**If yes, do the authors clearly describe all the elements that are included in the statistical testing procedure?** *This includes but is not limited to:* • *Two-sided or one-sided* • *Covariates* • *Robust standard errors* • *Two-stage procedure* ✓ Yes *If yes, please describe all the elements that are included in the statistical testing procedure.* For H1, we will use the gender attitude normalized score as the dependent variable (i.e., variable dep_var_h1) and the daughter dummy variable as the independent variable (i.e., variable d_daughter). Since we have a censored dependent variable, we will run a Tobit regression censored at 0 and 1, and will report a one-sided Romano-Wolf corrected for four hypotheses p-value. For H2, we will use the political preference variable as the dependent variable (i.e., variable dep_var_h2) and the daughter dummy variable as the independent variable (i.e., variable d_daughter). Since we have a censored dependent variable, we will run a Tobit regression censored at 0 and 10, and will report a one-sided Romano-Wolf corrected for four hypotheses p-value. For H3, we will use the share of women-specific stereotypical activities done by men within the household as the dependent variable (i.e., variable dep_var_h3) and the daughter dummy variable as the independent variable (i.e., variable d_daughter). Since we have a censored dependent variable, we will run a Tobit regression censored at 0 and 1, and will report a one-sided Romano-Wolf corrected for four hypotheses p-value. For H4, we will use the gender attitude normalized score as the dependent variable (i.e., variable dep_var_h4) and the daughter dummy variable interacted with a gender dummy variable as the independent variable (int_sex_daughter). Since we have a censored dependent variable, we will run a Tobit regression censored at 0 and 1 and will report a one-sided Romano-Wolf corrected for four hypotheses p-value.

		O No *If no, researchers do not describe all the elements that are included in the statistical testing procedure.* Justification:

<table>
<tr><td colspan="3">Multiple Hypothesis Testing (MHT)
Researchers should provide a correction method for multiple testing. This should include the correction method (e.g., Family-Wise Error Rate or False Discovery Rate) and a correction level justification according to the planned statistical testing. In case the researchers decide to include a MHT correction procedure or modify their pre-registered correction procedure ex-post, they should report it transparently. While adding an MHT correction ex-post can increase the severity of the testing, this should still be reported as a deviation. For more elements on deviations, please refer to the template “Deviations from the pre-registration”.</td></tr>
<tr><td>Are the researchers planning to test multiple hypotheses or run multiple statistical tests?</td><td>✓ Yes
If yes is selected, see elements in right column.

O No
If no is selected, researchers do not plan to test multiple hypotheses or run multiple statistical tests.
Justification:</td><td>If yes, do the researchers describe which correction method is chosen or justify why they do not correct?
Researchers should provide the chosen correction (e.g., Family-Wise Error Rate or False Discovery Rate).
For the Family-Wise Error Rate, this can include, but is not limited to, the following:
Bonferroni correction Dunn(2012)
Holm-Bonferroni correction Holm(1979)
Dependence List et al.(2019)
Romano-Wolf correction Romano and Wolf(2010) ; Clarke and Romano(2020); Arpinon and Espinosa(2023)
For the False Discovery Rate, the main resource is Benjamini and Hochberg(1995)

✓ Yes
If yes, please describe the chosen MHT correction method.
We will use a Romano-Wolf MHT correction for four hypotheses and report one-sided p-values.

O No
If no, researchers do not describe the chosen MHT correction method.
Justification:</td></tr>
</table>

		If an MHT correction method is chosen, do the researchers provide a justification for the correction level? *Note of caution: Researchers should provide a pre-defined alpha level (e.g., 5%) and clearly state how many hypotheses/statistical tests they will correct for. Omitting this step increases the analysis space and decreases the severity of the testing. Researchers should also include, if applicable, the chosen correction method in their power analysis.* ✓ Yes *If yes, please provide a justification for the correction level.* We will correct the one-sided estimated p-values for four hypotheses using a Romano-Wolf correction. The alpha level is defined by the #Manydaughters team. O No *If no, researchers do not provide a justification for the correction level.* Justification:

Multiverse analysis / Specification curves

Recent developments in open science have seen the emergence of multiverse analyses and specification curves. These tools have been developed to consider results' stability conditional on researchers' arbitrary choices during the data analysis. In a multiverse analysis or specification curve, researchers define each possible choice made during the research process (e.g., combining exclusion rule A with statistical model B and primary outcome definition A) and run each independent analysis on the same outcome. In the end, researchers are left with an array of results for the same outcome variable, allowing to verify the stability of a finding conditional on research choices. While in a pre-registration, researchers define the most plausible research path to analyze the data based on ex-ante beliefs, multiverse analyses and specification curves can be a useful complement. Here, researchers can decide to pre-register the multiverse analysis or specification curve, or they can simply include it as an exploratory analysis.

Do the researchers include a multiverse analysis or specification curve? *In the previous sections, researchers should have provided enough elements and justifications to argue why the chosen research path (i.e., analysis to be performed) is the most plausible one. However, sometimes researchers can be indifferent between two closely related branches and a multiverse or specification curve analysis can be performed.* *Some useful resources include Steegen et al.(2016) ; Wagenmakers et al.(2022) ; Girardi et al.(WP) ; Simonsohn et al.(2020).*	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers do not plan to run a multiverse or specification curve analysis.* Justification: The project #ManyDaughters does not allow for multiverse or specification curve analyses.	**If yes, do the researchers provide the complete number of possible combinations or universes they will explore?** O Yes *If yes, please describe the complete number of possible combinations or universes:* O No *If no, researchers do not describe the complete number of possible combinations or universes.* Justification:

Reliability checks/Outcome-neutral tests

Researchers should anticipate quality, or reliability checks for their measures to ensure validity. Researchers are free to pre-define the checks and the severity level for each check. For example, researchers can pre-register to change the statistical specification depending on the main variable of interest level (e.g., above or below a predefined threshold). Researchers can also set outcome-neutral tests and decide whether to perform a statistical analysis according to a pre-defined rule. If the researchers decide to add quality/reliability tests and outcome-neutral tests ex-post, this should be transparently reported. While adding an outcome-neutral test ex-post can increase the severity of the testing, it should still be reported as deviation. For more elements on deviations, please refer to the section in the "Deviations from the pre-registration".

Do the researchers provide quality/reliability check(s) to ensure the validity of their measures?	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers do not plan to use quality/reliability check(s) to ensure the validity of their measures.* Justification:	**If yes, do the researchers clearly describe how the quality/reliability check(s) will affect their analysis plan?** *Researchers should attempt to map out different data scenarios and how this could affect their statistical testing. Researchers can pre-register to change their statistical specification depending on how their main variable of interest is distributed or decide to censor if too many values are above/below a threshold.* *Example: researchers plan to access data in which the main variable is censored at 0 and 100, but are unsure how many observations are censored. If a limited number of observations are at the floor or ceiling, researchers commit to use an OLS regression. However, if more than 30% of observations are clustered at the floor or ceiling, researchers commit to use a Tobit model. This example is taken from Espinosa et al.(2024).* O Yes *If yes, please describe how the quality/reliability check(s):* O No *If no, researchers do not describe the quality/reliability check(s).* Justification:

Do the researchers provide outcome-neutral test(s)?	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers do not plan to use outcome neutral tests.* Justification:	**If yes, do the researchers clearly describe how each outcome-neutral test is defined?** *Researchers should provide a detailed description of the outcome neutral test(s) and the pre-defined procedure. This can include, but is not limited to, the following:* • *Variable threshold (e.g., values or below threshold)* • *Absence floor or ceiling effects* • *Positive controls Dienes(2020)* • *Set on ex-post power analysis Arpinon and Espinosa(2023)* *Note of caution: Researchers should also describe how the failure to pass an outcome-neutral test will impact the rest of the analysis plan. This is especially relevant for MHT correction and statistical testing.* O Yes *If yes, please describe the outcome neutral test(s).* O No *If no, researchers do not describe the outcome neutral test(s).* Justification:

Exploratory Analyses *Researchers might anticipate some additional analyses that are not central to their research project but arise as a second set of results. In this case, they have the choice to include or not these analyses in the pre-registration. Including them in the pre-registration increases transparency and decreases the analysis space. For this set of results, the researchers can decrease the level of severity (e.g., relax the Multiple Hypothesis Testing correction) and clearly identify the results as exploratory in the final manuscript. However, it is not always possible to anticipate all possible analytical paths. In case the researchers decide to run unplanned analyses, the results should be clearly reported in the final manuscript as unplanned exploratory results, upon which no strong conclusions are drawn. Again, researchers are free to adjust the severity of the testing but should not attempt to pass this set of results as predictions nor conclusive evidence.*		
Do the researchers specify any additional analyses that they are planning to run?	O Yes *If yes is selected, see elements in right column.* ✓ No *If no is selected, researchers do not plan to run additional analyses or will simply leave them for the exploratory analysis.* Justification: The project #ManyDaughters does not allow for exploratory analyses.	**If yes, do the authors discuss the additional analyses and how they plan to adjust the severity of testing for this set of results?** O Yes *If yes, please describe the additional analyses and adjusted severity of testing.* O No *If no, researchers do not describe the additional analyses and adjusted severity of testing.* Justification:

Other project specific elements *Researchers may be using unique features that do not fit in any of the Analysis Plan categories. Researchers should describe such features as precisely as possible. Omitting to do so may cause confusion and be interpreted as an increase in analysis space or multiverse.*	
Do the researchers have any additional features that do not fit in the categories above?	O Yes *If yes, please describe the additional features.* ✓ No *If no, researchers do not have any additional features.*